

Parent Functions I: $\sqrt{x}$ and $|x|$ **TEACHER KEY** | Algebra II Honors | Unit 1 | Day 4 | TEK A2.2.A**Objective:** I can graph and analyze the square root and absolute value parent functions, write absolute value as a piecewise rule, and explain why the two behave differently for negative inputs.**Vocabulary****Reflectional symmetry** — the graph can be **folded** across a line and land exactly on itself.**x-intercept(s)** — where the graph crosses the x-axis, where $y = 0$. **y-intercept(s)** — where $x = 0$.**Zeros** — the x-value(s) that make a function equal **0** — the same locations as the x-intercepts.**Parent function** — the simplest member of a family, with nothing added or multiplied.**Piecewise rule** — one function defined by **different expressions** on different parts of the domain.**Principal square root** — $\sqrt{a}$ means the **non-negative** root only, which is why $\sqrt{x}$ is a function.**Part 1 — Build Both Tables***Fill both. Note where outputs are undefined and where they repeat.*

x	-4	-1	0	1	4	9
$f(x) = \sqrt{x}$	undef.	undef.	0	1	2	3

x	-4	-1	0	1	4	9
$g(x) = x$	4	1	0	1	4	9

One table has undefined entries, the other has repeats. Explain what each tells you:

 $\sqrt{x}$ has a restricted **domain**; $|x|$ is not **one-to-one**, which will matter on Day 12.**Part 2 — Absolute Value as a Piecewise Rule**

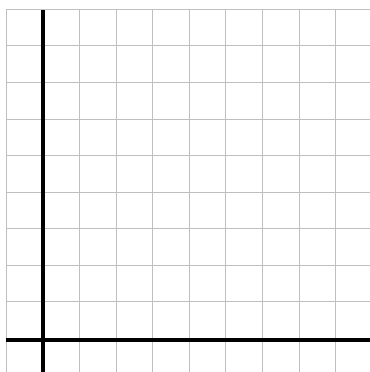
Absolute value is distance from zero, so it never returns a negative. Written as two cases:

$$|x| = x \text{ when } x \geq 0, \quad \text{and} \quad |x| = -x \text{ when } x < 0.$$

The second case looks like it produces a negative. Explain why it does not:

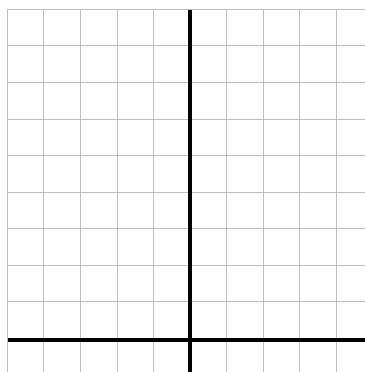
If x is already negative, **$-x$ is positive**. For example $-(-5) = 5$.**Part 3 — Graph Them Both***Each square is 1 unit. Plot from your tables, then connect smoothly.*

A. $f(x) = \sqrt{x}$ x from 0 to 9



Domain $[0, \infty)$ Range $[0, \infty)$

B. $g(x) = |x|$ x from -5 to 5



Domain **all reals** Range $[0, \infty)$

Part 4 — Solution Counts

Solve each. The number of answers is the point, not just the answers.

1. $|x| = 6 \rightarrow x = \mathbf{6 \text{ and } -6}$ how many? **2** 2. $\sqrt{x} = 6 \rightarrow x = \mathbf{36}$ how many? **1**

3. $|x| = 0 \rightarrow x = \mathbf{0}$ how many? **1** 4. $|x| = -3 \rightarrow \mathbf{\text{no solution}}$

Item 4 has no solution. What does that say about the range of $|x|$? **the output is never negative**

Connect item 1 to the graph: why do two x -values give the same output?

The V is symmetric about the **y-axis**, so a horizontal line meets it twice.

Part 5 — Going Further

1. Give the domain of each, in interval notation.

$\sqrt{x-7}$ **$[7, \infty)$** $\sqrt{3x+6}$ **$[-2, \infty)$** $|x-7|$ **$(-\infty, \infty)$**

2. Why does the absolute value one have no restriction while the radicals do?

3. On the interval $[1, 9]$, give the minimum and maximum of $\sqrt{x}$. min **1** max **3**

4. Is $\sqrt{x^2}$ the same function as $|x|$? Test $x = -5$ and $x = 5$, then decide.

Yes — **both give 5 for either input; $\sqrt{x^2} = |x|$ for every real x**

Exit Ticket

Both parent functions have minimum 0 and pass through the origin.

Name one attribute that separates them: **the domain** and one they share besides those two: **range $[0, \infty)$**

Write $|x|$ as a piecewise rule from memory: **x if $x \geq 0$, $-x$ if $x < 0$**